

Lab Assignment # 7.2

Program : B. Tech (CSE)
Specialization
Course Title : AI Assisted Coding
Course Code : 23CS002PC304
Semester : 11
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Batch No. : 52
Date : 30/01/26

Submission Starts here

Screenshots:

Task 1 — Runtime Error Due to Invalid Input Type

The issue is fixed by converting the input to a numeric type using float0 and handling invalid input with try-except

(Buggy Code):

```
num =  
input("Enter a  
number: ")
```

```
result = num +  
10
```

```
print(result)
```

```

ai_coding.py > ...
1  #Task 1 - Runtime Error Due to Invalid Input Type
2  #* A Python program accepts user input and performs arithmetic operations. However, the program throws a runtime error because the input is treated
3  #Example (Buggy Code):
4  #result = num + 10
5  #print(result)
6  #* Task:
7  #Use AI tools to identify the cause of the runtime error and modify the program so it executes correctly.
8  #Expected Output -1:
9  #* AI converts the input to the appropriate numeric type and eliminates the runtime error.
10 # Corrected Code
11 num = input("Enter a number: ")
12 num = float(num) # Convert input to a numeric type (float)
13 result = num + 10
14 print(result)
15 # Expected Output -2:
16 #* When the user inputs a number, the program correctly adds 10 and displays the result without any errors.
17 # Example Input: 5
18 # Example Output: 15.0
19 # Example Input: 5.5
20 # Example Output: 15.5
21
22

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\Eshwar\OneDrive\Desktop\python> & "C:\Program Files\Python314\python.exe" c:/Users/Eshwar/OneDrive/Desktop/python/ai_coding.py
Enter a number: 5
15.0
PS C:\Users\Eshwar\OneDrive\Desktop\python>

```

Task 2 - Incorrect Function Return Value

A function is designed to calculate the square of a number, but it does not return the computed result properly.

Example (Buggy Code):

```
def square(n):
    result = n * n
```

Task:

Use AI assistance to analyze the function and ensure the correct value is returned.

```

ai_coding.py > ...
1 #Incorrect Function Return Value
2 #A function is designed to calculate the square of a number, but it does not return the computed result properly.
3 #Example (Buggy Code):
4 # def square(n):
5 #     result = n * n
6 #Task:
7 # Use AI assistance to analyze the function and ensure the correct value is returned.
8 def square(n):
9     result = n * n
10    return result # Ensure the computed result is returned properly
11 # Example usage:
12 print(square(4)) # Output should be 16
13 print(square(5)) # Output should be 25
14 print(square(6)) # Output should be 36
15 # Corrected Code:
16 def square(n):
17     result = n * n
18     return result
19 # Ensure the computed result is returned properly
20 # Example usage:
21 print(square(4)) # Output should be 16
22 print(square(5)) # Output should be 25
23 print(square(6)) # Output should be 36

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\Eshwar\OneDrive\Desktop\python> & "C:\Program Files\Python314\python.exe" c:/Users/Eshwar/OneDrive/Desktop/python/ai_coding.py
16
25
36
16
25
36
PS C:\Users\Eshwar\OneDrive\Desktop\python>

```

Task 3- – IndexError in List Traversal

A Python program iterates over a list using incorrect index limits, causing an IndexError.

Example (Buggy Code):

```

numbers = [10, 20, 30]
for i in range(0, len(numbers)+1):
    print(numbers[i])

```

Task:

Use AI to identify the incorrect loop boundary and correct the iteration logic.

```
ai_coding.py > ...
1  #IndexError in List Traversal
2  #A Python program iterates over a list using incorrect index limits, causing an IndexError.
3  #Example (Buggy Code):
4  # numbers = [10, 20, 30]
5  # for i in range(0, len(numbers)+1):
6  #     print(numbers[i])
7  # Task:
8  # Use AI to identify the incorrect loop boundary and correct the iteration logic.
9  numbers = [10, 20, 30]
10 for i in range(0, len(numbers)):
11     print(numbers[i])
12 # Corrected Code:
13 # The loop now correctly iterates from 0 to len(numbers)-1, preventing IndexError.
14
15
16
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Eshwar\OneDrive\Desktop\python> & "C:\Program Files\Python314\python.exe" c:/Users/Eshwar/OneDrive/Desktop/python/ai_cod
10
20
30
PS C:\Users\Eshwar\OneDrive\Desktop\python> |
```

Task 4 – Uninitialized Variable Usage

A program uses a variable in a calculation before assigning it any value.

Example (Buggy Code):

if True:

pass

print(total)

Task:

Use AI tools to detect the uninitialized variable and correct the program.

```
ai_coding.py > ...
1 #Task 4 - Uninitialized Variable Usage
2 #A program uses a variable in a calculation before assigning it any value.
3 #Example (Buggy Code):
4 #if True:
5 #    pass
6 #    print(total)
7 #Task:
8 #Use AI tools to detect the uninitialized variable and correct the program.
9 #Corrected Code:
10 if True:
11     total = 0 # Initialize the variable before using it
12     print(total)
13
14
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Eshwar\OneDrive\Desktop\python> & "C:\Program Files\Python314\python.exe" c:/Users/Eshwar/OneDrive/Desktop/python/ai_coding.py
0
PS C:\Users\Eshwar\OneDrive\Desktop\python> |
```

Task 5 – Logical Error in Student Grading System

A grading program assigns incorrect grades due to improper conditional logic.

Example (Buggy Code):

```
marks = 85
```

```
if marks >= 90:
```

```
    grade = "A"
```

```
elif marks >= 80:
```

```
    grade = "C"
```

```
else:
```

```
    grade = "B"
```

```
print(grade)
```

Task:

Use AI to analyze the grading conditions and correct the logical flow.

```
ai_coding.py > ...
1  #Task 5 - Logical Error in Student Grading System
2  #A grading program assigns incorrect grades due to improper conditional logic.
3  #Example (Buggy Code):
4  #marks = 85
5  #if marks >= 90:
6  |   grade = "A"
7  #elif marks >= 80:
8  |   grade = "C"
9  #else:
10 |   grade = "B"
11 #print(grade)
12 #Task:
13 #Use AI to analyze the grading conditions and correct the logical flow.
14 marks = 85
15 if marks >= 90:
16 |   grade = "A"
17 elif marks >= 80:
18 |   grade = "B"
19 else:
20 |   grade = "C"
21 print(grade)
22
23
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS C:\Users\Eshwar\OneDrive\Desktop\python> & "C:\Program Files\Python314\python.exe" c:/Users/Eshwar/OneDrive/Desktop/python/ai_coding.py
B
PS C:\Users\Eshwar\OneDrive\Desktop\python> |
```